Syllabus

Class Schedule: Lecture: M W F: 11:00 AM – 11:50 AM (Section 02)

Date: Aug. 26, 2026 – Dec. 15, 2026

In-person: Integrated Science Center 3348

Instructor:
Yanhai Xiong
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Office Hours (in-person): W F 1 PM – 2 PM, or by appointment.

Course Description: Machine learning is the science of automating decisions and inferences based on data. It uses interdisciplinary concepts and techniques such as statistics, linear algebra, optimization, and computer science to create systems that can parse large amounts of data at high speeds and make automated predictions and decisions with or without human supervision. Machine learning is remarkably pervasive, with applications ranging from business intelligence to homeland security, medical analytics to biochemical engineering, environmental science to astrophysics, etc. This course will introduce students to a broad set of machine-learning ideas, models, and algorithms.

Course Objectives:

1. Provide students with a critical understanding of the essential aspects and techniques in machine learning and the use of data.

2. Develop students' ability to apply different machine learning methods and interpret the results.

3. Expose students to real-world problems that are being engaged with by contemporary problem-solvers and decision-makers.

Evaluation:

Class Participation	10%
Quizzes & Projects	50%
Midterm	20%
Final	20%

Letter Grades

93 – 100 %	A	73 – 76.99%	C
90 – 92.99%	A-	70 – 72.99%	C-
87 – 89.99%	B+	67 – 69.99%	D+
83 – 86.99%	B	63 – 66.99%	D
80 – 82.99%	B-	60 – 62.99%	D-
77 – 79.99%	C+	0.0 – 59.99%	F

Resources: For programming, you need access to reliable cloud computing, e.g, Google COLAB (https://colab.research.google.com/); you are recommended to use W&M Jupyterhub resource (https://jupyterhub.wm.edu); we use **PollEverywhere** (https://www.polleverywhere.com/home) for class discussions and participation; additional discussions are supported by **Blackboard**'s discussion function.
I aim to deliver the study material in a self-content and efficient manner, so no expensive textbook is necessary; in this course, you will engage with fundamental, conceptual ideas and applications.

Class Participation: You earn participation credit by joining class discussions regularly on PollEV. Your score matches your participation rate.

Quizzes & Projects: Quizzes will be *on paper in-class*, and involve short-answer, multiple-choice, and/or multiple answer questions, as well as explaining code provided to you (optionally using logical diagrams) to determine the answer. You will not be allowed a calculator on any quizzes because you will not need a calculator. You will be allowed to bring both sides of one *handwritten* A4 size notecard for each quiz with any notes that you want. You may *not print* the content of your notecard out for any quiz. Quizzes will not be cumulative. Quiz dates will be announced on Blackboard and via email *at least one day* in advance.

Projects assignments will be posted on Blackboard *at least 7 days* before the due date. The expectation is that you will work on the assignment as the material is covered, so do not be surprised if the due dates are relatively close to when we finish covering the material. If you have any questions, please post on Blackboard → Discussion. For projects, you will write your own code for solving a machine learning problem and get feedback on improving.

The grading of quizzes and projects also involves partial credit. The *lowest score from the Quizzes & Projects* section will be dropped to accommodate an unexpected short-term illness or class absence.

Midterm and Final: A midterm (Oct. 7th) and a final (Dec. 4th) will include both *in-class, closed-book, conceptual questions* (structured like a longer and cumulative quiz) as well as *take-home coding components* (like a project). They assess your understanding of the foundational concepts and machine learning algorithms covered in the course. You will not be allowed a calculator on any in-person exams because you will not need a calculator. You will be allowed to bring both sides of one handwritten piece of A4 paper with any notes that you want. You *may not print* the content of your notesheet out for the midterm or the final. We will collect your notesheet after the test.

Important Dates: Please refer the tentative course calendar and plan (including *important dates, weekly schedule etc.*) accordingly. This schedule is subject to change.

Course Schedule:

Module 1	Python Programming Review, Probability, Monte Carlo Simulations
Module 2	Confidence Intervals and Reinforcement Learning
Module 3	Regression/Classification Problems, Support Vector Machines
Module 4	Intro to Neural Networks (with PyTorch) and Deep Learning
Module 5	Convolutional Neural Networks and Grid Search Optimization Algorithms
Module 6	Natural Language Processing

Textbooks (Optional): *The Hundred-page Machine Learning Book* – Andriy Burkov
Data Science from Scratch: First Principles with Python – Joel Grus
Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems – Aurelien Geron
Machine Learning, A Probabilistic Perspective – Kevin P. Murphy

Late Policy & Extensions: Projects are typically due at 10:00 PM ET. Submissions made after the deadline will incur an immediate 10% penalty *of the total possible project value*. An additional 10% deduction of the total possible value will apply for every 24-hour period the assignment is late. *No assignments will be accepted more than 7 days past the original due date to ensure timely in-class feedback.*

You may request an extension before the deadline for any reason. If approved, late penalties will only apply after your new deadline. However, the absolute cutoff remains 7 days after the original due date. For extensions involving illness or emergencies, please provide supporting documentation (such as a doctor's note) within one week of approval.

- Example: For a project worth 100 max points where a student earns a base score of 80:
  - Submitted 15 hours late: Final score is 70 (80 minus a 10-point penalty).
  - Submitted 25 hours late: Final score is 60 (80 minus a 20-point penalty).

Collaboration: Collaboration is both allowed and encouraged! Most of the time in life, we are not working in isolation, and using the resources available is both wise and efficient. You are welcome to work collaboratively with classmates – that means you can work together to solve problems together…not that one person does the work and others copy it. You are also welcome to use the internet as a resource to refresh your memory, clarify concepts, and help with short code snippets. You must cite your sources, and you should not be copying anything wholesale from any source: internet, human, artificial intelligence, or otherwise.

Things that are off limits:

- Soliciting (hey, what did you get for question X? Can you send me your code?) and/or copying code and/or answers from another person
- Copying code/answers directly from internet sources
- Using sources appropriately but failing to cite them
- Getting AI to do your work for you

Think about it: do you want to do all of the work and then give the answers to someone? Is that fair to you? Do you wholeheartedly trust that the answers you get from sources (ahem, ChatGPT) will be 100% correct? Do you want to show up for a job interview and explain that you don't understand the material? Do you understand answers you are deriving from external sources well enough that you could explain them when asked?
You should always be ready and able to defend every answer you submit confidently - this means:

If I ask you to meet with me to explain any answer you submit, you should be able to do so in detail yourself.

Guidance: Students performing at a C level or below may schedule a meeting with the instructor to discuss class performance.

Class Absences: If you miss a class or two, check the material covered in class posted on Blackboard. For additional help, you can make an appointment with the instructor. Long-term absences need to be documented via the Dean of Students Office.

Honor Code: Academic integrity is at the heart of the university, and we all are responsible for upholding the ideals of honor and integrity. The student-led honor system is responsible for resolving any suspected violations of the Honor Code, and I will report all suspected instances of academic dishonesty to the honor system. The Student Handbook (www.wm.edu/studenthandbook) includes your responsibilities as a student. Your full participation and observance of the Honor Code are expected. To read the Honor Code, see www.wm.edu/honor.

Student Accessibility: William & Mary accommodates students with disabilities by federal laws and university policy. Any student who feels they may need an accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services staff at 757-221-2512 or sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation. For more information, please see www.wm.edu/sas.

Student Health: William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, and material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386.
- For additional support or resources & questions, contact the Dean of Students.
- For other resources available to students, see https://tinyurl.com/wmmentalhealth

Students who experience COVID-19 symptoms during the semester must make an appointment with the Student Health Center for a clinical assessment and testing if necessary. Students who test positive after the start of the semester and need to be quarantined must be cleared by the Student Health Center or a physician before returning to their residence hall and classes.

Living off campus? Visit the Off-Campus Isolation Guide.

The CDC calls for any COVID-19-positive person to isolate for at least five days.
Visit Quarantine & Isolation Guide and Calculator for more information.

Students who test positive should complete their five-day isolation at home or off campus if that is feasible. W&M does not coordinate dedicated COVID-19 quarantine or isolation housing.

In extenuating circumstances, residential students who cannot return home must be isolated in their room for five days, attending class remotely if possible and if they feel well enough. Otherwise, they should partner with professors to arrange to be out sick for their isolation period. Visit Academic Resources & Support.

For more information, visit:

https://www.wm.edu/about/administration/emergency/current_issues/coronavirus/index.php

For psychological/emotional stress, contact the Counseling Center by email or 757-221-3620.
For physical/medical concerns, contact the Health Center by email or at 757-221-4386.
Both are located at 240 Gooch Drive. Services are free and confidential.
If you or someone you know needs additional support or resources, please get in touch with the Dean of Students by submitting a care report online, by phone at 757-221-2510, or by email.